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Suhana Bedi

Department of Biomedical Data Science
Stanford University
453 Quarry Rd
Palo Alto, CA 94304, United States

E-mail: suhana@stanford.edu
Phone: (469) 931-4069
Linkedin: [linkedin.com/in/suhana-bedi-486a551b5](https://www.linkedin.com/in/suhana-bedi-486a551b5)
Google Scholar: [Google Scholar Profile](#)

RESEARCH INTERESTS

I am a second-year PhD student developing trustworthy foundation models for healthcare applications. My research creates rigorous evaluation frameworks for multimodal health AI systems that integrate electronic health records, clinical text, and medical imaging. I focus on three critical areas: assessing model robustness across deployment contexts, enabling temporal reasoning over longitudinal data, and ensuring reliable system integration of multi-component AI workflows. Through systematic benchmarking frameworks like MedHELM and automated failure discovery systems, I develop evaluation-guided adaptation strategies that bridge the gap between comprehensive assessment and targeted model improvements for safe deployment in clinical settings.

EDUCATION

Doctor of Philosophy, Biomedical Data Science Sep 2023 – Sep 2028
Stanford University
Co-Advisors: Dr. Nigam Shah and Dr. Sanmi Koyejo
GPA - 4.07

Bachelor of Science, Data Science Aug 2019 - May 2023
University of Texas at Dallas
GPA - 4.00
Summa Cum Laude

WORK AND TEACHING EXPERIENCE

Research Intern Jun 2025 – Sept 2025
Microsoft, Mountain View, CA

- Developed first longitudinal multimodal foundation model integrating EHR, CT imaging, and histopathology data through curriculum learning, achieving superior performance on 5 adverse event prediction tasks for oncology patients
- Conducted first multimodal patient trajectory scaling law study, establishing foundational insights for healthcare AI development
- Collaborated with Providence Health System to train and evaluate the model on a large-scale dataset of 2.3 million patient records, ensuring clinical relevance and data quality standards

Teaching Assistant Sep 2024 – Dec 2024
Stanford University, Palo Alto, CA

- BIOMEDIN 215: Data Science in Medicine
- Independently led an advanced technical lecture on Large Language Models, covering transformer architectures, attention mechanisms, and the complete LLM pipeline from pre-training to deployment. Deep-dived into transformer variants (GPT, BERT, T5), fine-tuning approaches, and alignment techniques like RLHF

- Mentored 70+ students in developing robust SQL queries for large-scale medical databases, focusing on data preprocessing, ETL pipelines, and implementing machine learning models

Student Researcher

Jan 2022 – Aug 2023

Google, Mountain View, CA

- Developed a biomedical knowledge graph integrating (300K+ entities, 950K triples) integrating 25+ healthcare datasets; implemented custom embeddings and entity linking, boosting model performance
- Built retrieval-augmented LLM system for clinical data access, reducing clinician query time and increasing dataset utilization

Research Intern

Jun 2021 – Dec 2021

Institute for Systems Biology, Seattle, WA

- Analyzed multi-omics datasets (genomics, transcriptomics, proteomics) using ML classification models to identify predictive biomarkers for pregnancy outcomes
- Implemented feature selection and correlation analysis pipelines to discover significant patterns in high-dimensional omics data

Research Assistant

Jan 2021 – Jun 2021

University of Texas at Dallas, Dallas, TX

- Benchmarked and compared normalization methods for spatial transcriptomics data; developed R package for streamlined gene expression analysis
- Applied clustering algorithms and dimensionality reduction techniques (PCA, UMAP) for automated cell-type identification in spatial data

Research Assistant

Aug 2019 – Jun 2021

Functional Genomics Lab, Dallas, TX

- Implemented ML methods in Visual Basic for phylogenetic analysis of interferons; built automated pipeline for variant and gene fusion detection across neurological databases
- Developed integrated analysis workflow combining RNA-seq and ChIP-seq data to identify key genomic signatures

PUBLICATIONS

⁺**First/Co-first author**

1. Fidelity of Medical Reasoning in Large Language Models

Suhana Bedi⁺, Yixing Jiang, Philip Chung, et al.

JAMA Network Open, 2025

Evaluated medical reasoning capabilities of 6 frontier LLMs using a "None of the Other Answers" (NOTA) methodology on MedQA questions, revealing substantial performance degradation (8.82%-38.24% accuracy drops) when models were forced to reason beyond familiar answer patterns, challenging claims of AI readiness for autonomous clinical deployment.

2. MedHELM: Holistic Evaluation of Large Language Models for Medical Tasks

Suhana Bedi⁺, Hejie Cui ⁺, Miguel Fuentes ⁺, Alyssa Unell ⁺, ..., Eric Horvitz, Percy Liang, Mike Pfeffer, Nigam Shah

Under Review

Introduced the first comprehensive benchmark suite for real-world medical LLM evaluation, featuring a clinician-validated taxonomy of 121 tasks across 5 categories and 35 benchmarks (18 newly created), revealing significant performance gaps between licensing exam scores and actual clinical capabilities across 9 frontier models.

3. **[Oral Presentation] The Optimization Paradox in Clinical AI Multi-Agent Systems**
Suhana Bedi⁺, Iddah Mlauzi, Daniel Shin, Sanmi Koyejo, Nigam H. Shah
ACM KDD Workshop on Evaluation and Trustworthiness of Agentic and Generative AI Models, 2025
 Revealed an important paradox in clinical AI multi-agent systems: better individual components (85.5% information accuracy) led to worse patient outcomes (67.7% vs. 77.4% diagnostic accuracy), demonstrating that system-wide integration matters more than component optimization in healthcare AI deployment.
4. **Fidelity of Medical Reasoning in Large Language Models**
Suhana Bedi⁺, Yixing Jiang, Philip Chung, Sanmi Koyejo, Nigam H. Shah
JAMA Network Open, 2025
 Proposed a robustness evaluation method that exposes LLMs' reliance on pattern matching over reasoning in medicine. When correct answers were replaced with "None of the other answers", all 6 frontier models showed significant accuracy drops, challenging readiness claims for autonomous clinical deployment and highlighting the need for robustness-focused evaluation beyond standard benchmarks.
5. **Measurement to Meaning: A Validity-Centered Framework for AI Evaluation**
 Olawale Salaudeen⁺, Anka Reuel⁺, Ahmed Ahmed, **Suhana Bedi**, Zachary Robertson, Sudharsan Sundar, Ben Domingue, Angelina Wang. Sanmi Koyejo
Under Review
 Introduced a psychometrics-based validity framework to bridge the gap between AI model performance on narrow benchmarks and broader capability claims, providing structured methodology for evaluating whether test performance indicates specific task proficiency or general reasoning abilities.
6. **Enabling Accessible Biomarker Prediction Through 3D Body Mesh Estimation from Surface Scans**
 Ayis Pyrros⁺, Muhammad Ahmed Chaudhry⁺, **Suhana Bedi**, Pola Lydia Lagari, *et al.*
MLHC 2025 Clinical Abstract Track, 2025
 Developed machine learning models that predict key imaging biomarkers from consumer-grade surface scans using 3D body meshes, achieving promising accuracy compared to traditional CT/MRI methods.
7. **Rethinking Evaluation of Large Language Models in Healthcare**
Suhana Bedi⁺, Nigam Shah, Sanmi Koyejo
CPI Tech Chronicles, 2025
 Proposed a consequential validity-driven evaluation framework for LLMs in healthcare, addressing gaps in conventional NLP benchmarks by integrating real-world clinical impact, workflow integration, and policy implications.
8. **Distilling Large Language Models for Efficient Clinical Information Extraction**
 Karthik S. Vedula⁺, Annika Gupta⁺, Akshay Swaminathan⁺, Ivan Lopez, **Suhana Bedi**, Nigam Shah
Under Review
 Demonstrated distilled BERT models achieve 101x lower costs and 12x faster inference than LLMs, with comparable clinical NER performance.
9. **Context Clues: Evaluating Long Context Models for Clinical Prediction Tasks on EHR Data**
 Michael Wornow⁺, **Suhana Bedi**⁺, Miguel Hernandez, Ethan Steinberg, Jason Fries, Chris Ré, Sanmi Koyejo, Nigam Shah
International Conference on Learning Representations (ICLR), 2025
 Conducted first systematic evaluation of long-context Foundation Models on EHR data, showing significant performance gains across clinical tasks.
10. **ZIP-FIT: Embedding-Free Data Selection via Compression-Based Alignment**
 Elyas Obbad⁺, Iddah Mlauzi, Brando Miranda, Rylan Schaeffer, Kamal Obbad, **Suhana Bedi**, Sanmi Koyejo
ICML Workshop on Unifying Data Curation Frameworks Across Domains, 2025
 Developed novel compression-based method for efficient training data selection without embedding computations.

11. **meds_reader: A fast and efficient EHR processing library**
 Ethan Steinberg⁺, Michael Wornow⁺, **Suhana Bedi**⁺, Jason Fries, Matthew McDermott, Nigam Shah
Machine Learning for Health Symposium (ML4H), 2024
 Built optimized Python package achieving 10-100x efficiency improvements in EHR data processing.
12. **[Oral Presentation] QUEST-AI: A System for Question Generation, Verification, and Refinement using AI for USMLE-Style Exams**
Suhana Bedi⁺, Scott Fleming⁺, Chia-Chun Chiang, Keith Morse, Aswathi Kumar, Birju Patel, Jenelle Jindal, Conor Davenport, Craig Yamaguchi, Nigam Shah
Pacific Symposium on Biocomputing (PSB), 2025
 Developed system for automated generation and validation of high-quality USMLE-style medical exam questions.
13. **[Editor's Choice] Testing and Evaluation of Health Care Applications of Large Language Models**
Suhana Bedi⁺, Yutong Liu⁺, Lucy Orr Ewing⁺, Dev Dash, Sanmi Koyejo, Alison Callahan, Jason Fries, Michael Wornow, Akshay Swaminathan, Lisa Lehmann, Hyo Jung Hong, Mehr Kashyap, Akash Chaurasia, Nirav Shah, Karandeep Singh, Troy Tazbaz, Arnold Milstein, Michael Pfeffer, Nigam Shah
Journal of the American Medical Association (JAMA), 2024
 Led largest systematic review of 519 healthcare LLM studies, identifying critical gaps in real-world evaluation and deployment.
14. **Feasibility of Automatically Detecting Practice of Race-Based Medicine by Large Language Models**
 Akshay Swaminathan⁺, Sid Salvi, Philip Chung, Alison Callahan, **Suhana Bedi**, Alyssa Unell, Mehr Kashyap, Roxana Daneshjou, Nigam Shah, Dev Dash
AAAI 2024 Spring Symposium on Clinical Foundation Models
 Created automated methods to detect and analyze race-based medicine in clinical AI systems.
15. **Spatial Transcriptomics Arena (STAr): an Integrated Platform for Spatial Transcriptomics Methodology Research**
 Xi Jiang⁺, Danni Luo, Esteban Fernández, Jie Yang, Huimin Li, Kevin W Jin, Yuanchun Zhan, Bo Yao, **Suhana Bedi**, Guanghua Xiao, Xiaowei Zhan, Qiwei Li, Yang Xie
bioRxiv, 2023
 Built computational platform for efficient spatial transcriptomics data analysis and visualization.
16. **Similarities between bacterial GAD and human GAD65: Implications in gut mediated autoimmune type 1 diabetes**
Suhana Bedi⁺, Tiffany M Richardson⁺, Baofeng Jia⁺, Hadeel Saab, Fiona SL Brinkman, Monica Westley
PLoS One, 2022
 Investigated molecular similarities between bacterial and human proteins to understand autoimmune diabetes mechanisms.

AWARDS AND HONORS

1. William R. Hewlett Fellowship, Stanford University 2023 - 2025
 - Highly selective fellowship awarded to outstanding PhD students in Computer Science
2. Academic Excellence Scholarship, University of Texas at Dallas 2019 - 2023
 - Full-tuition merit scholarship awarded to top 1% of incoming undergraduates

TALKS

1. “Evaluating Healthcare LLMs: From Benchmarks to Real-World Advances” (**Doctoral Symposium**), *AHLI CHIL 2025, Berkeley, CA, 2025*
2. “Difficult Methodological Challenges in Evaluating AI in Health” *Google Health, Mountain View, 2025*
3. “Evaluating Large Language Models with frameworks and tools for healthcare” *BrainX, 2025*
4. “Large Language Models in Healthcare: A Technical Deep Dive” (**Guest Lecture: Data driven medicine**), *Stanford University, 2024*
5. “LLMs for Operational Efficiency and Enhancing Clinical Support Systems (**AI + Health Conference**)”, *Stanford University, 2024*
6. “Testing and Evaluation of Health Care Applications of Large Language Models (**Healthy ML Group**)”, *Massachusetts Institute of Technology, 2024*

SERVICE (Journal Reviewer)

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| • PLOS Medicine | 2025 |
| • Journal of the American Medical Informatics Association (JAMA) Open | 2025 |
| • New England Journal of Medicine AI | 2024 |
| • Nature Medicine | 2024 |